Backtests as Reward Signals: Controlled False Discovery in Autonomous Alpha Research

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ABSTRACT

Autonomous research systems can propose trading rules, observe their backtests, and condition the next proposal on the observed score. This turns a validation backtest into a reward signal and makes the tested family itself data-adaptive. We measure the resulting false-discovery channel under exact global nulls. The confirmatory synthetic benchmark contains 2,000 heavy-tailed stochastic- volatility markets, 1.5 million candidate evaluations, and three matched search policies. At a budget of 250, target feedback adds 0.255 validation-Sharpe units (95% CI 0.246 to 0.265) relative to independent sham feedback, but adds 0.000 on a sealed confirmation sample (95% CI -0.020 to 0.021). Results replicate on 300 sign-randomized paths constructed from five canonical Kenneth French portfolio panels: the cell-level effect is 0.222 (95% CI 0.192 to 0.253) in validation and -0.010 (95% CI -0.072 to 0.052) in confirmation. Raw-count Deflated Sharpe Ratio (DSR) is conservative in this grammar, whereas a participation-ratio trial-count heuristic, Bonferroni, and BH remain feedback-sensitive; target false-promotion rates are 42.0%, 9.7%, and 35.2%. A single-use sealed test remains calibrated at 5.30%. A four-arm Claude research agent with exact, sham, blind, and coarsened feedback produces an exact-minus-sham validation contrast of 0.181 and a confirmation contrast of 0.027. A planted-signal study exposes the corresponding power costs. The contribution is a falsification protocol and causal feedback contrast, not a claim that any particular correction fails universally or that the tested strategies earn deployable alpha.

KEYWORDS

backtest overfitting, adaptive data analysis, alpha mining, multiple testing, large language model agents, quantitative finance

1 INTRODUCTION

The marginal cost of generating and backtesting a trading rule is approaching zero. Recent systems combine large language models (LLMs), evolutionary search, tree search, code execution, and backtest feedback to automate parts of factor research [11, 12, 15, 16]. Benchmarks now measure formula generation, implementation, and end-to-end strategy design [5, 13, 18]. Throughput is useful, but it changes the statistical object: candidate $j + 1$ may be a function of scores returned for candidates $1, \ldots, j$ on the same history.

Finance already treats data snooping as a first-order risk. Reality checks, superior-predictive-ability tests, the probability of backtest overfitting, the Deflated Sharpe Ratio (DSR), and higher factor-discovery thresholds address different parts of this problem [1, 2, 9, 10, 17]. These tools remain important. Our question is narrower: what happens when the definitions of the observed strategies are adapted to the same score later used as evidence?

This question is difficult to answer in historical markets. Failure after discovery could be selection bias, alpha decay, costs, or a regime change; success could be skill, a risk premium, or luck. We therefore start with ground-truth nulls. We compare an adaptive search rewarded on the target validation path with the same search rewarded on an independent sham path. Their initial candidates, grammar, stochastic decisions, and budget are matched. Both ultimately select the best target-validation candidate. The target-minus-sham contrast isolates the incremental effect of letting the proposal mechanism observe the scored noise.

The paper makes five contributions.

(1) It gives a simple Gaussian counterexample showing why a correction based only on the realized tested family cannot generally restore validity when the hypothesis definitions depend on the scored sample.
(2) It estimates the feedback effect on 2,000 fresh synthetic null markets, rather than relying on the earlier 200-market design pilot.
(3) It constructs exact conditional nulls from five widely used Kenneth French daily panels and three eras, preserving each realized cross-sectional second moment while eliminating predictable signed returns.
(4) It replaces the hand-coded proposer with a recorded Claude agent and randomizes the information channel across exact, sham, blind, and coarsened feedback.
(5) It reports a planted-signal experiment so a firewall is evaluated on both false-positive size and power.

The claims are deliberately bounded. We do not show that DSR, Bonferroni, or BH is invalid in every intended setting. We show that, in these controlled feedback loops, corrections computed over executed trials can be sensitive to how the trials were generated. Nor do we treat sign-randomized historical panels as evidence of profitable strategies.

2 CLOSEST WORK AND THE MISSING TEST

2.1 Financial data snooping

White's Reality Check evaluates whether the best model in a specified universe has genuine predictive superiority [17]; Hansen's SPA test improves power by reducing the influence of poor alternatives [9]. DSR compares a selected Sharpe with an expected-maximum hurdle and adjusts its sampling distribution for non-normal returns [2]. Harvey, Liu, and Zhu emphasize that the economically relevant multiplicity includes a research literature, not only tests printed in one paper [10]. Our design makes that usually hidden research path experimentally manipulable.

2.2 Adaptive validation

Adaptive data analysis shows that repeatedly querying a validation sample can destroy classical generalization guarantees. A reusable holdout controls the information released from a protected sample [6]; the Ladder releases a score only after a sufficiently large improvement [4]. An automated backtester is a financial leaderboard whose submissions are executable hypotheses. We adapt the sham-feedback idea into a paired causal experiment and test a coarse Ladder-style channel in the Claude arm.

2.3 Autonomous factor research

AlphaAgent, AlphaJungle, R&D-Agent-Quant, and related systems explicitly use financial evaluation to guide later generations [11, 15, 16]. AlphaBench evaluates generation, search, and validation; AlphaForgeBench reports run-to-run instability in end-to-end design [13, 18]. These are complementary to our study. We do not attempt a broad model leaderboard. We test Type-I error under a known null, randomize whether a proposal policy receives target information, and open a temporally disjoint confirmation sample once.

To our knowledge, the combination of an exact global null, paired target-versus- sham feedback intervention, sealed temporal confirmation, and promotion-gate calibration has not been reported for autonomous alpha research. The novelty is this controlled falsification harness, not the general observation that researchers can overfit.

3 WHY COUNTING OBSERVED TRIALS IS NOT ENOUGH

Let $X \in \mathbb{R}^d$ represent standardized validation noise. For a fixed unit vector u, the strategy statistic $Z_u = u^\top X$ is standard normal and the one-sided p-value $1 - \Phi(Z_u)$ is valid. The next proposition isolates the issue created by an adaptive proposal definition.

Proposition 1 (A realized-family counterexample). *Let $0 < \alpha < 1/2$ and $X \sim \mathcal{N}_d(0, I_d)$ with $d \geq 2$. An adaptive researcher observes X, defines the single reported direction $\widehat{u} = X/\|X\|_2$, and evaluates it as though it were fixed. Then the reported statistic is $Z_{\widehat{u}} = \|X\|_2$. A nominal one-sided level-α fixed-direction test rejects with probability*

$$\Pr(\chi_d > z_{1-\alpha}) > \alpha,$$

and that probability converges to one as $d \to \infty$. The realized candidate count and its participation-ratio effective count are both one.

Proof. For every non-random unit vector u, rotational invariance gives $u^\top X \sim \mathcal{N}(0,1)$. For the data-dependent direction, $\widehat{u}^\top X = \|X\|_2 \sim \chi_d$. At the positive threshold $z = z_{1-\alpha}$, $\|X\|_2 \geq |X_1|$ implies $\Pr(\chi_d > z) \geq 2[1 - \Phi(z)] = 2\alpha > \alpha$. Moreover, $\|X\|_2/\sqrt{d} \to 1$ in probability while $z_{1-\alpha}$ is fixed, so the rejection probability converges to one. □

The construction is intentionally stark. It does not say that a particular DSR formula behaves this way in our bounded grammar. It says that realized family size cannot, by itself, encode the information used to define hypotheses. A valid correction must either account for the complete selection policy, limit the information returned by the protected data, or evaluate the selected strategy on information the policy never saw.

This distinction matters for interpreting the results. We report two DSR variants. Canonical DSR uses the raw executed-trial count N. A sensitivity substitutes the participation ratio

$$N_{\text{eff}} = \frac{N^2}{\sum_{i,j} R_{ij}^2}, \tag{1}$$

where R is the trial-return correlation matrix. The latter is a heuristic effective dimension, not a theorem for adaptively selected trials. A difference between feedback arms diagnoses the information channel; a rule's overall distance from 5% diagnoses its calibration in this particular benchmark.

4 EXPERIMENTAL PROTOCOL

4.1 Prospective status and experimental units

The earlier 200-market exercise was used as a design pilot. Before the deep runs, we fixed an internal prospective analysis plan dated 25 August 2026, canonicalized behaviorally duplicate specifications, separated raw-count and effective-count DSR columns, strengthened the no-look-ahead test, fixed fresh seed ranges, and committed to reporting reversals and null results. This was not an external preregistration. The 2,000-market confirmation begins at seed 50,000; pilot rows are not pooled with it.

The independent synthetic market is the unit for Experiment A. In the empirical randomization experiment, the 20 sign paths within a dataset–era cell share a realized return panel; inference therefore averages paths first and treats the 15 dataset–era cells as the primary units. Claude's four repeated quartets per cell are aggregated the same way.

4.2 Synthetic global null

For each seed we generate 30 asset returns. With seed-specific loadings B, common Student-t_6 factors f_t, idiosyncratic Student-t_6 shocks e_t, and persistent log volatility h_t, returns are

$$r_t = 0.012 \exp(h_t) D^{-1} \left(B f_t + \operatorname{diag}(s) e_t\right), \tag{2}$$

$$h_t = 0.975 h_{t-1} + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, 0.1^2). \tag{3}$$

The normalizer D gives each asset comparable unconditional scale. The process is heavy-tailed, heteroskedastic, and cross-sectionally correlated, but $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = 0$. A market contains 252 warm-up sessions, 1,260 scored validation sessions, a 252-session gap that also warms the confirmation backtest, and 1,260 scored confirmation sessions.

4.3 Frozen strategy grammar and execution

For fast and slow lookbacks $L_f < L_s$, let $z_{f,t}$ and $z_{s,t}$ be cross-sectional standardized cumulative returns known through $t-1$. The score is

$$x_t = a z_{f,t} + (1 - |a|) z_{s,t}, \qquad a \in [-1, 1]. \tag{4}$$

The backtester buys the top k and sells the bottom k with unit gross and zero net exposure, using equal or inverse-volatility weights. Fast windows are $\{5, 10, 21, 42, 63\}$; slow windows are $\{21, 42, 63, 126, 252\}$; k is $\{2, 3, 5, 7\}$; and rebalancing is every

$\{1, 5, 10, 21\}$ sessions. An equal-weighted specification canonicalizes its irrelevant volatility window so semantically identical rules cannot inflate the trial count.

A signal through close $t-1$ earns the close-to-close return at t. Turnover is the L_1 change in target weights. The primary experiments set costs to zero because their estimand is Type-I error, not economic profitability. Tests verify that asset-specific future perturbations cannot alter earlier weights and do alter later weights.

4.4 Matched search policies

Candidate budgets are nested prefixes of 1, 10, 25, 50, 100, and 250.

Blind batch samples candidate parameters independently of backtest results, then selects the highest target-validation Sharpe. It captures ordinary data snooping.

Adaptive sham begins with five random strategies. Later proposals mutate an elite strategy with probability 0.8 and draw independently otherwise. Elites are ranked on an independent sham market. Target performance is logged but cannot affect proposal generation.

Adaptive target uses the same initial strategies, random decisions, mutation operator, and final target-selection rule, but ranks parents using the target validation scores. Its difference from adaptive sham is the primary feedback estimand.

4.5 Promotion rules

For every selected strategy we compute a one-sided Newey–West HAC test of positive mean return [14]. The naive rule uses $p < 0.05$. Bonferroni uses $p < 0.05/N$ for the selected strategy. BH reports whether any executed candidate is rejected at $q = 0.05$ [3]. Under the global null this event is a false discovery.

DSR compares the selected Sharpe with the expected maximum of N zero-skill trials and accounts for sample length, skewness, and kurtosis [2]. We label the raw-N implementation "DSR" and the participation-ratio substitution "DSR N_{eff}." A sealed rule requires one-sided confirmation HAC probability of at least 0.95. Dual rules require both a validation DSR and the sealed test. The confirmation frame is opened only after strategy selection.

Continuous paired contrasts receive Student-t intervals; synthetic promotion rates receive Wilson intervals. The empirical and Claude primary intervals use 15 cell averages and are conditional on the fixed panels, independently drawn sign paths, and (for Claude) independent model calls. Because the five panels share a market calendar and security universe, we also report a deliberately conservative sensitivity that first averages panels within each of the three non-overlapping eras. For synthetic confirmation we pre-specified practical equivalence when the 90% interval lies within $[-0.10, 0.10]$ Sharpe; this judgmental bound treats an annualized Sharpe difference of 0.10 as the smallest policy effect of practical concern.

Two-sided paired sign-flip tests enumerate all assignments for at most 20 units; larger samples use 99,999 deterministic Monte Carlo draws with the standard plus-one correction. This makes the p-value resolution explicit.

4.6 Exact nulls on canonical market panels

We use five daily, value-weighted families from the Kenneth French Data Library [7, 8]: 49 industries, 25 size–book-to-market portfolios, 100 size–book-to-market portfolios, 25 size–momentum portfolios, and 25 size–short-term-reversal portfolios. Three blocks begin in 1990, 2002, and 2014. Each has a one-year validation warm-up, five validation years, a one-year gap/confirmation warm-up, and five confirmation years.

For a fixed observed return cross-section r_t^{obs}, draw independent Rademacher signs ε_t and define

$$r_t^0 = \varepsilon_t r_t^{\text{obs}}, \qquad \Pr(\varepsilon_t = 1) = \Pr(\varepsilon_t = -1) = \tfrac{1}{2}. \tag{5}$$

This preserves $r_t^0 (r_t^0)^\top = r_t^{\text{obs}} (r_t^{\text{obs}})^\top$ exactly, as well as missingness, absolute returns, jumps, and the calendar. For any weights measurable from past signed returns,

$$\mathbb{E}_\varepsilon \left[w_t^\top r_t^0 \mid \varepsilon_1, \ldots, \varepsilon_{t-1}, R^{\text{obs}} \right] = 0. \tag{6}$$

Thus the null is finite-sample exact under the sign-randomization measure, conditional on the fixed historical panel. Target, sham, and confirmation use independent signs. Sharing the unsigned panel holds the risk regime fixed and makes target-minus-sham conservative for overfitting to preserved magnitudes, which cancels between arms. Twenty paths per cell produce 300 environments and 225,000 maximum-budget candidate evaluations.

4.7 Claude research agent

The agent experiment replaces the mutation operator with Claude Sonnet invoked through Claude Code CLI 2.1.241. Each of 60 matched quartets shares a target sign path, grammar, and five initial strategies. The design therefore contains 240 arm-level episodes. Batching ten episodes per request across five rounds and four call groups produces 120 archived model calls. Five rounds request four new strategies, for a budget of 25. The four arms are:

- *exact*: six-decimal Sharpe on the target path;
- *sham*: the same statistic on an independently signed path;
- *blind*: prior specifications with no numeric score;
- *coarsened*: a running-best target Sharpe rounded down to 0.10 and updated only when the next 0.10 bin is crossed.

Arms in the same quartet are sent in separate, concurrent calls. Opaque episode identifiers and randomized call order reveal neither arm, panel, nor era. The model never receives raw returns, asset labels, dates, confirmation results, or another arm's history. A strict JSON schema freezes the grammar. Invalid or duplicate proposals trigger a logged random fallback. Prompts, response wrappers, canonical model identifiers, latency, token use, cost, and fallbacks are archived.

Hosted-model generations are nondeterministic. The target-minus-sham estimate is therefore an intent-to-treat contrast between two stochastic search policies, not a within-generation counterfactual. Model sampling variability is part of the policy and is averaged across four quartets in each of 15 cells. Claims apply only to the recorded snapshot and protocol.

4.8 Planted-signal power

The alternative experiment adds a known 21-session cross-sectional momentum premium to the synthetic risk process. If u_t is the base null return and q_{t-1} is the cross-sectionally standardized 21-session sum of lagged u, then

$$r_t = u_t + \theta q_{t-1}, \tag{7}$$

for $\theta \in \{0.5, 1.0, 2.0\}$ basis points per day. The zero-strength function is bit-identical to the null generator. For each nonzero strength we run 250 fresh markets. Adaptive sham receives an independent development market with the same planted mechanism, so it measures portable discovery rather than target-noise exploitation. This is a controlled power calculation, not a claim about the prevalence or form of real alpha.

5 RESULTS

5.1 Synthetic confirmation: feedback increases validation optimism

Figure 1 shows selected validation and sealed confirmation Sharpe by candidate budget. At 250 candidates, target feedback adds 0.255 validation Sharpe relative to sham (95% CI 0.246 to 0.265; the paired sign-flip test attains its smallest reportable Monte Carlo value, $p = 1/(99{,}999 + 1) = 10^{-5}$). This reports the chosen resolution; it does not imply finer p-value precision. The corresponding confirmation difference is 0.000 (95% CI -0.020 to 0.021), and its 90% interval lies inside the pre-specified $[-0.10, 0.10]$ equivalence band. The optimism-gap contrast is therefore almost exactly the validation contrast. This is not merely selection from 250 fixed draws: blind and sham searches share multiple testing, but only target feedback can condition later definitions on target noise.

5.2 Promotion rules separate calibration from feedback sensitivity

Table 1 and Figure 2 report maximum-budget false promotions. Raw-count DSR is conservative: target promotes 0.00% and its Wilson upper bound is below 0.2%. This reverses the headline impression from the design pilot because the pilot stored the participation-ratio substitution in an ambiguously named DSR column. The corrected artifact keeps both columns.

The N_{eff} heuristic promotes 42.0% of target searches, versus 24.1% for sham and 12.5% for blind. Bonferroni promotes 9.7% of target searches but only 1.45–1.85% of controls. BH gives 35.2% in target and 2.75–3.05% in controls. Two conclusions should not be conflated. Their target–control ordering is evidence that the proposal feedback channel matters; their absolute rates describe these implementations under this grammar, not universal size theorems.

The sealed rule is different. It promotes 5.10% of blind, 5.05% of sham, and 5.30% of target searches. The feedback policy changes validation selection but cannot change the independent signs or shocks in confirmation.

5.3 The effect replicates on five canonical panels

The exact empirical null reproduces the feedback result across every panel family. The average target-minus-sham validation effect across 15 independently signed panel–era cells is 0.222 (conditional 95% CI 0.192 to 0.253); every panel-family mean is positive. Confirmation is -0.010 (95% CI -0.072 to 0.052). The cell-level paired sign-flip p-values are 6.1×10^{-5} for validation and 0.735 for confirmation. When panels are first averaged within each of only three non-overlapping eras, the validation 95% interval is 0.132 to 0.313. Target Bonferroni promotions are 9.33%, DSR N_{eff} promotions are 36.33%, and sealed promotions are 5.33%. Raw-count DSR promotes 0.00% of empirical target paths.

5.4 Claude treats backtests as an optimizable reward

At budget 25, the cell-averaged Claude exact-minus-sham validation contrast is 0.181 (conditional 95% CI 0.093 to 0.268; paired sign-flip $p = 0.0015$), while exact-minus-blind is 0.127 (95% CI 0.053 to 0.200). The coarsened-minus- exact contrast is -0.099 (95% CI -0.170 to -0.027). Exact-minus-sham confirmation is 0.027 (95% CI -0.078 to 0.131; $p = 0.594$), consistent with no confirmation difference. The validation sensitivity that first averages the five panels within each of three eras is 0.023 to 0.338. The resulting optimism-gap contrast is 0.154 (95% CI -0.008 to 0.316).

The agent result is interpreted as a policy effect, not as a statement about Claude in general. Separate calls prevent direct arm comparison but do not remove hosted-model sampling noise. Fallback rates and an analysis excluding fallback-contaminated episodes are necessary robustness checks because an arm with more schema failures partly becomes a random-search arm. Candidate-level fallback rates are 0.9% for exact target and 1.3% for sham. Restricting to the 36 of 60 quartets with no target- or sham-arm fallback attenuates the contrast to 0.108 (cell-level 95% CI 0.012 to 0.205; $p = 0.032$), but it remains positive. This is a robustness analysis, not a replacement for the intent-to-treat estimand.

5.5 Firewall power under planted alpha

Figure 5 reports promotion probability for target-adaptive search under controlled alternatives. Sealed confirmation pays for validity by using an independent five-year sample; raw-count DSR pays through an extremely high validation hurdle in this correlated grammar. Bonferroni and the dual N_{eff}-plus-sealed rule occupy different size–power points. The full policy table prevents the high apparent power of a mis-sized rule from being mistaken for useful discovery performance.

6 WHAT A RESEARCH FIREWALL MUST PROTECT

The experiments imply a systems property rather than a preferred scalar correction. The statistical boundary should be drawn around information flow.

6.1 Operational protocol

(1) Create three isolated capabilities: a development store with full access, a protected validation service, and a write-once confirmation service that the proposer cannot address directly.

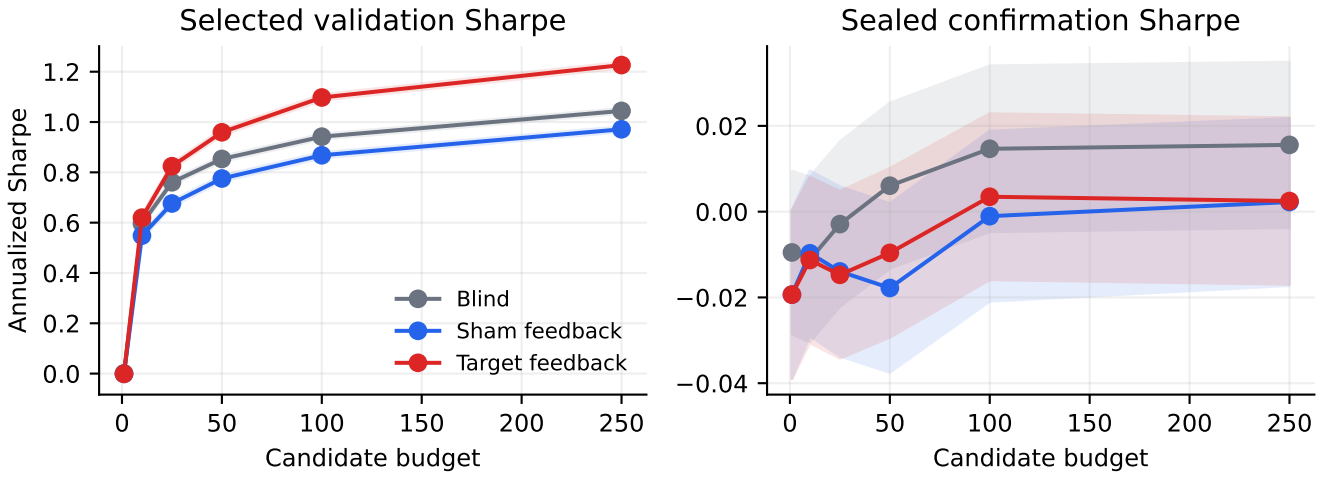


Figure 1: Mean selected Sharpe across 2,000 fresh synthetic null markets. Bands are descriptive 95% normal-approximation intervals around the mean. Smaller budgets are prefixes of the same maximum-budget trajectory.

Table 1: Synthetic false-promotion rates at budget 250 (percent). "Raw N" is canonical DSR; "N-eff" substitutes the participation ratio.

Policy	Naive	DSR raw N	DSR N-eff	Bonf.	BH	Sealed	N-eff + sealed
Blind	91.5	0.0	12.5	1.8	3.0	5.1	0.7
Sham feedback	82.2	0.1	24.1	1.5	2.8	5.1	1.0
Target feedback	97.2	0.0	42.0	9.7	35.2	5.3	2.2

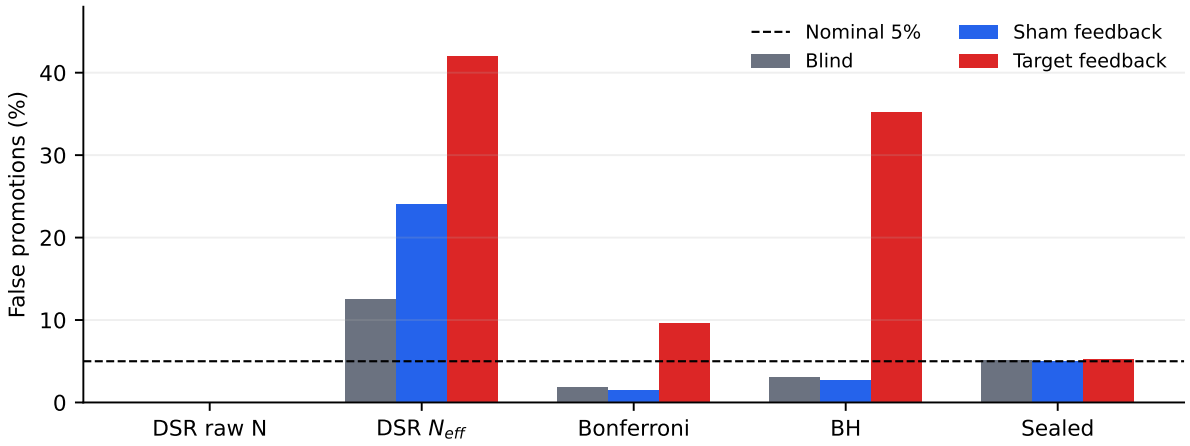


Figure 2: False-promotion rates in 2,000 synthetic global-null markets. The horizontal line is the nominal 5% level.

(2) Permit full development-to-proposer flow. If iterative validation is unavoidable, release only a pre-specified coarsened score. Permit exactly one confirmation-to-decision flow; never return confirmation data or scores to the proposer.
(3) Canonicalize and semantically deduplicate every executable strategy before it consumes the frozen trial budget.
(4) Append every prompt, model snapshot, tool call, retry, code repair, fallback, executable fingerprint, and score release to an immutable, hash-chained ledger.
(5) Freeze the budget, selection rule, promotion gates, and failure policy before the first protected query.

Table 2: Empirical exact-null results at budget 250. Differences are target minus sham; promotion columns are target-arm path rates.

Panel	Validation diff.	Confirmation diff.	Target Bonf. (%)	Target sealed (%)
49 industries	0.227	0.038	15.0	6.7
Size–BM 100	0.259	-0.050	10.0	3.3
Size–BM 25	0.207	-0.048	6.7	5.0
Size–momentum 25	0.204	-0.119	6.7	1.7
Size–reversal 25	0.215	0.130	8.3	10.0

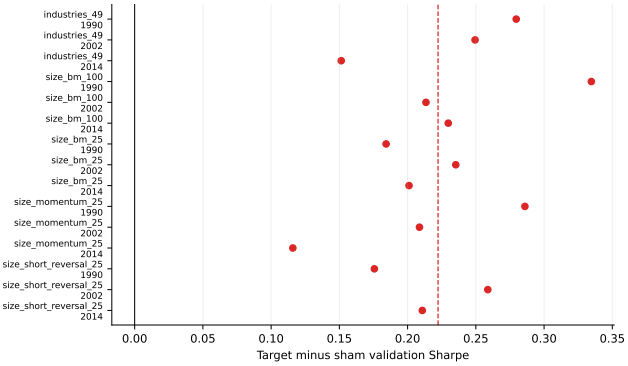


Figure 3: Target-minus-sham validation Sharpe for each panel–era cell, averaging its 20 sign-randomized paths. Dashed line: mean across 15 cells.

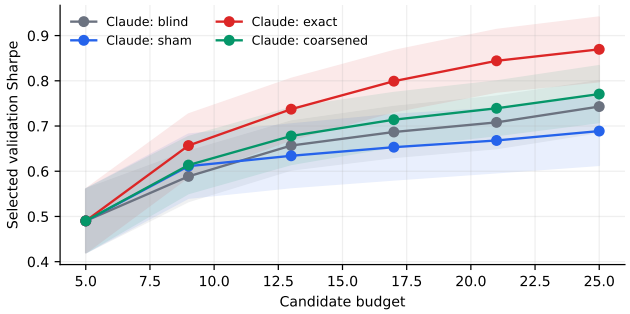


Figure 4: Claude selected validation Sharpe by feedback channel. Each point averages 60 episodes; intervals treat episodes descriptively, while reported inference uses 15 panel–era cell averages.

Table 3: Claude outcomes at budget 25. Promotion rates are descriptive over 60 episodes per arm.

Arm	Validation Sharpe	Confirmation Sharpe	Bonf. (%)	Sealed (%)
Blind	0.743	-0.030	1.7	0.0
Sham	0.689	-0.008	1.7	1.7
Exact	0.870	0.018	6.7	5.0
Coarsened	0.771	0.028	1.7	10.0

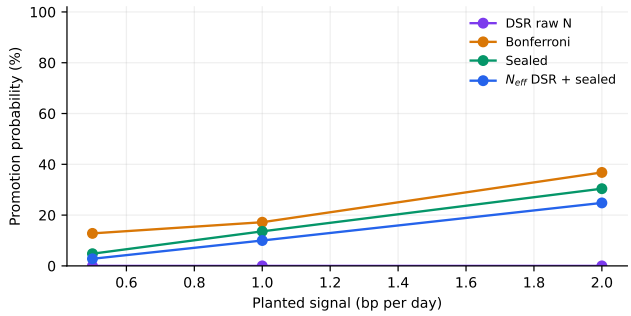


Figure 5: Promotion probability at budget 250 for target-adaptive search under planted cross-sectional momentum. Null sizes are reported separately in Table 1; curves should be compared at matched size.

(6) Select one strategy, query confirmation once, and promote only if its pre-specified validation and confirmation gates both pass.
(7) After unauthorized access, a missing log segment, or a second confirmation query, abort promotion, quarantine the candidate, burn the confirmation sample, and restart under a new frozen protocol. Never retune on the failed confirmation result.

Only the single-use confirmation step is directly calibrated by our experiments. Coarsening limits feedback bandwidth but is not a general validity guarantee, and semantic deduplication remains grammar-specific. The protocol is therefore an auditable engineering boundary, not a theorem that every permitted workflow controls false discovery.

Promotion rules should be calibrated under the complete search mechanism. A gate valid for a fixed universe need not remain valid after the scoring sample helps generate that universe; exact-null simulations should rerun the policy, not only resample returns for a frozen winner. Size and power must be reported together: an unreachable hurdle is not useful discovery, while a mis-sized rule's high discovery rate is not power. Governance belongs on a size–power–compute frontier.

For practitioners, the irreducible production controls are concrete: a write-once confirmation service the proposer cannot address, a hash-chained record of every prompt, score release, retry, and fallback, and a budget and promotion gate frozen before the first protected query. Exact-null calibration is an up-front cost rather than a per-search ritual. Once deployed, the most direct

Table 4: Planted-signal promotion probabilities at budget 250 (percent). Sham uses an independent market with the same planted mechanism.

Signal (bp/day)	Policy	DSR raw N (%)	Bonf. (%)	Sealed (%)	N-eff + sealed (%)
0.5	Blind	0.0	2.4	6.8	1.2
0.5	Sham feedback	0.0	3.2	6.0	2.4
0.5	Target feedback	0.0	12.8	4.8	2.8
1.0	Blind	0.0	8.4	11.6	2.4
1.0	Sham feedback	0.0	6.0	9.6	6.0
1.0	Target feedback	0.0	17.2	13.6	10.0
2.0	Blind	0.0	24.8	33.2	11.2
2.0	Sham feedback	0.0	23.6	31.2	22.0
2.0	Target feedback	0.0	36.8	30.4	24.8

diagnostic is the gap between selected validation and single-use confirmation performance: under a fixed budget, a widening gap is evidence that the search is learning the scoring sample rather than a transferable signal. Coarsening and semantic deduplication can reduce bandwidth, but their validity remains grammar-specific and must be re-calibrated rather than assumed.

7 LIMITATIONS

First, the frozen strategy language is small. That is useful for auditability but does not represent arbitrary code, alternative data, portfolio construction, or execution modeling. A richer language may change both overfitting speed and gate calibration.

Second, the synthetic markets and sign-randomized panels target false discovery under known nulls. Sign randomization conditions on a realized historical panel and destroys serial sign structure; it is not a model of the unconditional market distribution. Shared magnitudes make the target-minus-sham contrast a lower bound for overfitting components common to both arms.

Third, only one hosted Claude alias and recorded snapshot are used. Four quartets per cell reduce generation variance but cannot establish stable cross-model rankings. Provider-side updates limit exact reruns even when every prompt is archived.

Fourth, the coarsened arm is a simple Ladder-style mechanism, not a differentially private reusable holdout. It may reduce the bandwidth of leakage without guaranteeing validity. The sealed confirmation result is the cleanest positive control.

Fifth, transaction costs, survivorship, portfolio capacity, and nonstationary real alpha are not the paper's target. None of the null-panel results should be used to select a live strategy.

8 REPRODUCIBILITY AND DISCLOSURE

The artifact contains the frozen TOML protocol, source, 61 tests, 2,000-market candidate audit, empirical sign seeds, all raw-file SHA-256 hashes, Claude transcripts, model/cost metadata, result hashes, analysis script, generated tables, and figures. Across the four experiments it records 2,293,500 candidate evaluations. French archives are downloaded from the source during reproduction and are not redistributed. Smaller budgets are computed as prefixes of one maximum-budget search, preventing budget-specific reseeding.

We distinguish artifact reproduction from an experimental rerun. A one-command clean-environment check verifies every frozen-manifest source, transcript, and result input before analysis; runs the linter and unit tests; regenerates tables, macros, and figures in isolated staging; compares deterministic numeric content; and compiles this paper. It fails closed if a required hash, test, headline value, or text table differs. The synthetic, sign-randomized empirical, and planted-signal pipelines are seeded, but this release check recomputes their reported analyses from archived trials rather than rerunning every candidate search. The hosted Claude generations are not seed-reproducible, so agent statistics are recomputed from frozen transcripts and trials without querying the provider. Figure and paper PDF byte hashes are recorded only as provenance because toolchain metadata can vary. This verifies the internal computational consistency of the released artifact. A second implementation, which does not import the manuscript analysis, recomputes 23 tolerance-compared components directly from the three frozen trial CSVs, checks the pre-specified 90% equivalence band, and enforces the 2,000-market, 15-cell, three-era, and 15-cell agent aggregation units. This same-data cross-check can detect divergent coding or aggregation, but it does not rerun candidate searches, constitute independent replication, or detect errors shared by the archived trials and their generating code.

Claude was used in two roles. It was a measured proposal policy in the agent experiment (120 calls, $17.54 recorded service cost), where every interaction is archived, and a critical coauthoring assistant used to challenge the statistical interpretation. The human author fixed the protocol, inspected code and artifacts, resolved conflicts against the producing code version, and accepts responsibility for the paper. No LLM output is treated as empirical evidence unless reproduced by the frozen benchmark.

9 CONCLUSION

A backtest score is not passive when an autonomous researcher can act on it; it is a reward signal. Across 2,000 parametric null markets and exact conditional nulls built from five canonical portfolio panels, target-feedback search policies select higher validation Sharpe without higher sealed confirmation Sharpe. Corrections over executed trials react differently: raw-count DSR is very conservative, a participation-ratio substitution and BH are strongly anti-conservative, and Bonferroni remains feedback-sensitive. A single-use sealed test stays near nominal size.

The practical lesson is modest but consequential. Trustworthy autonomous alpha research requires a firewall around information flow, an audit of the complete search policy, and calibration under null experiments that rerun that policy. More agents and more backtests increase research capacity; without that firewall, they also increase the capacity to manufacture evidence from noise.

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A PRE-SPECIFIED DEVIATIONS AND CORRECTIONS

The deep revision differs from the pilot in four correctness-critical ways. First, strategies with equal weighting canonicalize the otherwise irrelevant volatility window, eliminating behavioral duplicates. Second, the artifact now stores raw-N DSR and participation-ratio DSR in separate columns. The pilot's 43.5% target figure belonged to the latter despite ambiguous naming; an exact recomputation of canonical raw-N DSR did not reproduce it. Third, the future- perturbation test changes an asset-specific cross-section rather than adding a common shock that a market-neutral book cancels. Fourth, all confirmatory seeds, data hashes, model identifiers, and output hashes are explicit. The analysis plan is honestly labeled an internal prospective plan, not an external preregistration.

Concurrent model calls are an implementation optimization added before the confirmatory Claude run. The four matched arms remain in separate calls, and no result was inspected before this change. Calls within a round are independent; subsequent rounds still wait for all four arms, preserving the frozen history.

B NEXT PAPERS AND GO/NO-GO TESTS

The benchmark supports, but does not automatically justify, a research program. The next methods paper should map the size–power frontier across signal strength, half-life, breadth, turnover, and regime decay; it proceeds only if a governance mechanism yields a useful Pareto improvement. A selection-aware randomization paper should rerun the full researcher on signed panels; it proceeds only if it controls size on independent null draws and improves data efficiency over a sealed holdout. An agent-audit paper should test whether prompt branches, retries, semantic duplicates, and trace description length predict sealed-sample decay beyond executed trial count. Finally, a contamination- resistant benchmark should distinguish rediscovery of published factors from discovery of randomly encoded planted mechanisms. The companion research program records designs and stopping rules for each.